CLPB Deficiency — 3-Methylglutaconic Aciduria Type VIIB (MEGCANN)

Disease: CLPB Deficiency 3-Methylglutaconic Aciduria Type VIIB **MONDO ID:** MONDO:0014561 · **OMIM:** #616271 · **Orphanet:** ORPHA:445038 / ORPHA:324631 **Category:** Mendelian (inborn error of mitochondrial protein quality control)

Summary

CLPB deficiency (3-methylglutaconic aciduria type VIIB; MEGCANN) is an ultra-rare, autosomal-recessive inborn error of mitochondrial protein quality control caused by variants in the *CLPB* gene (11q13.4), which encodes a HAX1-activated AAA+ disaggregase/refoldase resident in the mitochondrial intermembrane space (IMS). The protein's normal job is to disaggregate and refold stress-damaged proteins in the IMS; when its activity is lost, IMS client proteins (including HAX1 and HTRA2) aggregate and the SPY proteolytic quality-control complex (STOML2–PARL–YME1L1) is crippled. The clinical consequence is a severe multisystem syndrome combining **congenital neutropenia** (which can transform to MDS/leukemia), **intellectual disability**, **progressive brain atrophy**, **epilepsy**, **movement disorder (including dystonia)**, **cataracts**, and the biochemical hallmark **3-methylglutaconic aciduria (3-MGA)**.

The disease illustrates a striking genotype–phenotype architecture. **Biallelic loss-of-function** produces the full, often lethal, multisystem MEGCANN syndrome, whereas **monoallelic dominant-negative missense variants** — clustered near the ATP-binding pocket — cause a milder, neutropenia-predominant phenotype (frequently isolated severe congenital neutropenia, often without 3-MGA). This dual inheritance mechanism is explained structurally: the human CLPB ortholog Skd3 forms spiral hexamers that engage substrate via pore loops and stack head-to-head into dodecamers through adaptable ankyrin-repeat (ANK) interactions. SCN-linked subunits poison the hexamer (dominant-negative), while MGCA7-linked subunits do not. Consistent with this, gnomAD shows *CLPB* is **not haploinsufficient** (pLI ≈ 0), so monoallelic disease arises from protein poisoning rather than dosage loss.

Prognosis is poor for the severe biallelic form (~66% mortality; 50% four-year survival), with structural-impact predictions (nonsense-mediated decay, ankyrin-domain disruption) being stronger survival predictors than zygosity or age of onset. **No disease-specific or curative therapy exists.** Management is entirely supportive: G-CSF (filgrastim) for neutropenia with HSCT reserved for leukemic transformation, antiepileptics and rehabilitation for neurologic features, and cataract surgery. The 3-MGA biomarker itself is not treatable and is a fluctuant, sometimes-undetectable secondary metabolite rather than a disease driver.

1. Disease Information & Key Identifiers

Resource	Identifier
Gene symbol	**CLPB** ("ClpB family mitochondrial disaggregase")
NCBI Gene	81570
HGNC	HGNC:30664
Gene OMIM	616254
Ensembl	ENSG00000162129
UniProt	Q9H078
Cytogenetic locus	11q13.4 (GRCh38 chr11:72,285,495–72,434,680, minus strand)
Disease OMIM	**#616271** (MEGCANN / MGCA7B)
MONDO	**MONDO:0014561**
Orphanet	ORPHA:445038 (and ORPHA:324631, "3-methylglutaconic aciduria type 7")
DOID	DOID:0081134, DOID:0110003
GARD	0017767
MedGen / UMLS	C5676893
SNOMED CT	764860006

Synonyms: 3-methylglutaconic aciduria with cataracts, neurologic involvement, and neutropenia (MEGCANN); 3-methylglutaconic aciduria-cataract-neurologic involvement-neutropenia syndrome; MGA7; MGCA7; MGCA7B; "CLPB 3-methylglutaconic aciduria." Information in this report is derived from aggregated disease-level resources (OMIM, Orphanet, MONDO, gnomAD, ClinVar) and published patient cohorts, not from EHR/individual-patient data.

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2. Etiology

- **Primary cause:** monogenic (Mendelian). Biallelic pathogenic variants in *CLPB* cause the full MEGCANN syndrome; monoallelic dominant-negative missense variants cause a neutropenia-predominant presentation. There is no environmental, infectious, or multifactorial etiology.
- **Genetic risk factors:** the causal variants themselves (see Section 4); consanguinity increases the chance of biallelic genotypes in affected families.
- **Environmental risk/protective factors:** none established — the disease is genetically determined.
- **Gene–environment interactions:** not applicable; no documented modifier exposures.

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3. Phenotypes

Across cohorts, biallelic CLPB deficiency presents with a consistent constellation. Wortmann et al. (2021) summarized it: *"biallelic CLPB variants underlie a neurological phenotype ranging from nonprogressive intellectual disability to prenatal encephalopathy with progressive brain atrophy, movement disorder, cataracts, 3-methylglutaconic aciduria, and neutropenia."* The founding paper emphasized the severity range and malignancy risk: *"neutropenia that can develop into leukemia, a neurological phenotype ranging from nonprogressive intellectual disability to a prenatal encephalopathy with progressive brain atrophy, movement disorder, cataracts, and early death."* Neutropenia and intellectual disability are the hallmarks. [PMID: 34140661;

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Phenotype	HPO term	Type	Onset / severity / frequency
Congenital neutropenia (→ leukemia/MDS)	HP:0000844	Laboratory/ hematologic	Congenital; hallmark; can transform to malignancy
Intellectual disability	HP:0001249	Clinical sign	Hallmark; nonprogressive → severe
Developmental delay	HP:0001263	Clinical sign	Early childhood; common
Progressive brain atrophy	HP:0012444	Imaging/ structural	Progressive in severe forms
Seizures / epilepsy	HP:0001250	Clinical sign	Frequent in severe forms
Movement disorder / dystonia	HP:0001332	Clinical sign	Common
Cataracts	HP:0000518	Physical manifestation	Characteristic; surgically treatable
3-methylglutaconic aciduria	HP:0003535	Laboratory abnormality	Biochemical hallmark; fluctuant

Quality-of-life impact: severe in the biallelic form — early mortality, profound neurodisability, recurrent infections, visual impairment, and lifelong dependency. The monoallelic neutropenia-predominant form has a comparatively better functional trajectory. No disease-specific QoL instrument (EQ-5D/SF-36/PROMIS) data are published for this ultra-rare condition.

4. Genetic / Molecular Information

Causal gene: *CLPB* (NCBI Gene 81570; HGNC:30664; gene OMIM 616254), 11q13.4.

Variant spectrum (ClinVar, queried Oct 2025): ~954 *CLPB* records — ~157 pathogenic, ~38 likely pathogenic, ~692 of uncertain significance. Disease-causing classes: - **Missense:** biallelic p.Arg408Gly abolishing ATPase (P 25597510); SCN-associated heterozygous missense clustering near the ATP-binding pocket

(34115842).
- **Nonsense/frameshift:** stop/stop genotypes carry the highest disease burden (41719910).
- **Splice variants.**

All reported variants are **germline**; no somatic/cancer-driver role is established. **Functional consequence:** biallelic = loss of function (recessive); specific monoallelic missense = **dominant-negative**. In-silico structural impact (NMD, ankyrin-domain disruption) predicts survival (41719910).

Population constraint (gnomAD, Oct 2025): pLI ≈ 2.4×10^{-5} (not haploinsufficient); observed/expected LoF = 0.51 (40 observed vs 78.4 expected); LOEUF = 0.66; LoF Z = 3.68; missense Z = 1.53. Heterozygous LoF alleles occur in and are tolerated by the general population — explaining why a single null allele does **not** cause disease, whereas a single *poisoning* (dominant-negative) missense protein does. [PMID: 34115842]

Modifier genes / epigenetics / chromosomal abnormalities: no established modifier genes, epigenetic mechanisms, or large-scale chromosomal rearrangements for this disease; it is a single-gene disorder.

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5. Environmental Information

Not applicable. CLPB deficiency is a purely genetic disorder with no documented environmental, lifestyle, or infectious contributors.

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6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic (or dominant-negative monoallelic) pathogenic variant in *CLPB* (11q13.4) →** reduces or abolishes functional CLPB AAA+ disaggregase/refoldase protein in the mitochondrial intermembrane space. *(demonstrated —*

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2. **Loss of CLPB ATPase/refoldase activity** → CLPB can no longer disaggregate and refold stress-damaged IMS client proteins. Normal activation requires HAX1 binding the ankyrin domain to drive the active hexamer; this step fails. *(demonstrated —*

P 42244541

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P 36170828

3. **Loss of disaggregase activity → aggregation of IMS proteins** (CLPB, HAX1, HTRA2) under stress. *(demonstrated —*

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P 38270563

4. **IMS protein aggregation → impaired SPY complex (STOML2–PARL–YME1L1) function** → degraded mitochondrial protein quality control. *(demonstrated —*

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P 38270563

5. **Branch A (hematologic):** impaired IMS proteostasis + loss of the CLPB/HAX1 axis → impaired granulocytic differentiation and increased apoptosis of hematopoietic progenitors → congenital neutropenia → (with clonal pressure) risk of MDS/leukemia. *(demonstrated in cells —*

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P 34115842

6. **Branch B (neurologic):** mitochondrial dysfunction in CNS → progressive neuronal loss → cerebral/cerebellar atrophy, intellectual disability, epilepsy, movement disorder. *(model-supported —*

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inferred for human CNS)

7. **Branch C (ophthalmologic):** mitochondrial/proteostatic stress in the lens → cataracts. *(observed clinically; mechanism inferred)*
8. **Branch D (biochemical):** impaired mitochondrial energy metabolism → acetyl-CoA buildup → trans-3-methylglutaconyl-CoA → 3-methylglutaconic acid excretion (secondary, fluctuant

biomarker). *(mechanism inferred —*

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  CLPB variant (11q13.4)
         |
  loss of CLPB disaggregase/refoldase  <-- (needs HAX1 cofactor; fails)
         |
  aggregation of IMS proteins (CLPB, HAX1, HTRA2)
         |
  SPY complex (STOML2-PARL-YME1L1) dysfunction
         |
  ------------------------------------------------
  |            |             |             |
Hematologic  Neurologic    Ophthalmic    Metabolic
neutropenia  brain atrophy  cataracts    acetyl-CoA buildup
(-> leukemia) ID/epilepsy/             -> 3-MGA aciduria
              movement disorder          (fluctuant biomarker)
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Upstream vs downstream: the mutation and loss of disaggregase activity are the upstream lesions; IMS aggregation and SPY dysfunction are the proximate molecular consequences; neutropenia, brain atrophy, cataracts, and 3-MGA are downstream, tissue-level manifestations. 3-MGA is the most downstream and least specific — a secondary readout of mitochondrial energy failure rather than a driver.

Supporting detail

CLPB is *"a mitochondrial intermembrane space AAA+ domain-containing disaggregase"* (Baker et al., 2024). In CLPB-null cells, stress triggers aggregation specifically in the IMS and *"CLPB deficiency impeding SPY complex function by virtue of protein aggregation in the intermembrane space."* [PMID: 38270563] HAX1 is the direct activating cofactor: *"HAX1 is a direct stimulatory cofactor of CLPB. HAX1 promotes oligomerization of CLPB into an active disaggregase and stimulates the ATPase and refoldase activities of the oligomeric complex."* [PMID: 42244541] On the hematologic branch, *"Both genetic loss of CLPB and expression of CLPB variants result in impaired granulocytic differentiation of human hematopoietic progenitor cells and increased apoptosis."* [PMID: 34115842]

3-MGA biochemistry: CLPB deficiency is a **secondary** 3-MGA syndrome (impaired mitochondrial energy metabolism/membrane integrity), unlike primary MGCA1 (*AUH*, leucine catabolism). *"this accumulation is due to an underlying buildup of acetyl-CoA in the mitochondria from impaired function of the TCA cycle with ensuing formation of trans-3-methylglutaconyl CoA and its subsequent byproducts, including 3-MGA."* [PMID: 39787888]

Ontology anchors: - GO biological process: GO:0042026 (protein refolding), GO:0051131 (chaperone-mediated protein complex assembly), GO:0006457 (protein folding). - GO cellular component: **GO:0005758 (mitochondrial intermembrane space)**, GO:0005739 (mitochondrion). - CL cell types: CL:0000775 (neutrophil), CL:0000037 (hematopoietic stem cell), CL:0000540 (neuron), CL:0000121 (Purkinje/cerebellar neuron). - UBERON: UBERON:0000955 (brain), UBERON:0002037 (cerebellum), UBERON:0000965 (lens), UBERON:0002371 (bone marrow). - CHEBI: 3-methylglutaconic acid; CHEBI:15351 (acetyl-CoA).

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7. Anatomical Structures Affected

- **Primary organs / systems:** central nervous system (cerebrum and cerebellum; UBERON:0000955, UBERON:0002037), hematopoietic/immune system (bone marrow, neutrophil lineage; UBERON:0002371), eye/lens (UBERON:0000965).
- **Secondary involvement:** systemic consequences of neutropenia (recurrent infections).
- **Subcellular compartment: mitochondrial intermembrane space** (GO:0005758) is the defining site of molecular pathology.
- **Lateralization:** brain atrophy typically bilateral/symmetric; cataracts bilateral.

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8. Temporal Development

- **Onset:** congenital/neonatal for the severe biallelic form (can be prenatal encephalopathy); neutropenia is congenital. Milder forms present in childhood.
- **Course:** progressive brain atrophy and neurodisability in severe forms; neutropenia is chronic with risk of leukemic transformation.
- **Critical window:** onset before 12 months of age predicts more severe disease (P 41719910).

- **3-MGA timing:** fluctuant and may appear only with delay in neonates — a diagnostic pitfall (P 31729175).

9. Inheritance & Population

- **Inheritance:** autosomal recessive (biallelic) for the full MEGCANN syndrome; **monoallelic de novo / dominant-negative** missense variants cause a neutropenia-predominant presentation.
- **Penetrance/expressivity:** highly variable expressivity (nonprogressive ID → lethal neonatal encephalopathy); severity tracks with variant class (NMD/ankyrin-disruption worst).
- **Constraint:** *CLPB* is LoF-tolerant in heterozygotes (pLI ≈ 0), confirming recessive biology plus a separate dominant-negative route.
- **Prevalence:** ultra-rare; Orphanet lists prevalence as unknown/<1:1,000,000; ~60–100 individuals reported worldwide (largest cohort = 63 patients, P 41719910).
- **Demographics:** pan-ethnic; consanguinity reported in some families; no established founder population; both sexes affected without strong sex bias.

10. Diagnostics

- **Biochemical:** urine organic acids showing elevated 3-MGA (fluctuant, may be absent/delayed — repeat testing advised); CBC showing neutropenia. Untargeted plasma metabolomics is an emerging, more sensitive screen (P 39787888).
- **Hematologic:** absolute neutrophil count; bone marrow examination (maturation-arrest pattern typical of SCN); surveillance for MDS/leukemia.
- **Imaging:** brain MRI showing cerebral/cerebellar atrophy (progressive).

- **Genetic testing (confirmatory, gold standard):** WES/WGS or targeted panels (mitochondrial/3-MGA/SCN panels including *CLPB*); single-gene *CLPB* sequencing when the phenotype is recognized. ClinVar/ACMG classification; structural-impact prediction (NMD, ankyrin-domain disruption) adds prognostic value (P 41719910).
- **Differential diagnosis:** other 3-MGA syndromes — Barth (*TAZ*, MGCA2), *OPA3* (MGCA3/Costeff), *DNAJC19* (MGCA5/DCMA), *HTRA2* (MGCA8), *YME1L1*, primary MGCA1 (*AUH*); other causes of severe congenital neutropenia (*ELANE*, *HAX1*, *G6PC3*).
- **Screening:** carrier/cascade testing in affected families; no population newborn screening exists for this disorder specifically.

11. Outcome / Prognosis

Measure	Biallelic	Monoallelic
Mortality	66%	23%
4-year survival	50%	82%
Burden by genotype	Stop/stop > missense combinations	Dominant-negative missense, milder
Strongest survival predictor	Structural impact (NMD, ANK disruption) > zygosity or onset age	—

Onset before 12 months predicts more severe disease. Complications: life-threatening infection (from neutropenia), leukemic/MDS transformation, intractable epilepsy, progressive neurodisability, visual impairment from cataracts. [PMID: 41719910]

12. Treatment (NCIT anchors)

No disease-specific or curative therapy exists; care is multidisciplinary and symptom-directed. The neutropenia follows the SCN paradigm: *"Granulocyte colony-stimulating factor and hematopoietic stem cell transplantation are now the bedrock of standard care."* [PMID: 36485107] In mechanistically related HAX1-CN, *"All patients were initially treated with G-CSF with a sufficient increase in absolute neutrophil counts."* [PMID: 37193639]

- **G-CSF / filgrastim** (NCIT:C1512) — first-line for neutropenia.
- **Hematopoietic stem cell transplantation** (NCIT:C15431) — for leukemic transformation/ refractory neutropenia.
- **Antiepileptic drugs** (NCIT:C264) — seizure control.
- **Cataract surgery** — lens opacity.
- **Rehabilitation** — PT/OT/speech therapy for neurodevelopmental support.

No gene therapy, RNA-based therapy, targeted small molecule, or pharmacogenomic guidance exists for this disease. 3-MGA itself is not specifically treatable. [PMID: 31030818]

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13. Prevention

Primary prevention is limited to genetic counseling and reproductive options (carrier testing in at-risk families, prenatal/preimplantation genetic testing) given the Mendelian basis. Secondary prevention centers on early neutrophil monitoring and infection prophylaxis once diagnosed; tertiary prevention is leukemia surveillance and supportive multidisciplinary care. No vaccine or environmental intervention is applicable.

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14. Other Species / Natural Disease

- **Evolutionary conservation:** CLPB orthologs exist across metazoa — *"The multicellular metazoan lineage acquired a novel chaperone in the mitochondrial intermembrane space."* [PMID: 42244541]
- **Natural veterinary disease:** none documented; relevance is as a research model, not a companion-animal disease.

- **Zoonotic potential:** not applicable (genetic disorder).

15. Model Organisms

- **Zebrafish (*Danio rerio*) *clpb* morphant** — established in-vivo model: morpholino suppression produced a CNS phenotype with cerebellar/cerebral atrophy, rescued by WT but not mutant human *CLPB* mRNA. *"Suppression of clpb in zebrafish embryos induced a central nervous system phenotype that was consistent with cerebellar and cerebral atrophy that could be rescued by wild-type, but not mutant, human CLPB mRNA."* [PMID: 25597510] This demonstrates conserved neurodevelopmental requirement and loss-of-function causality.
- **Cellular/in-vitro models:** patient-derived fibroblasts (complexome profiling showing CLPB/HAX1 high-mass comigration, P 34140661); human hematopoietic progenitors and cell lines showing impaired granulocytic differentiation and increased apoptosis (P 34115842); CLPB-null cell lines showing IMS aggregation (P 38270563); reconstituted recombinant Skd3/CLPB disaggregase assays (P 36170828, P 42244541).
- **Genetic model types available:** morpholino knockdown (zebrafish), gene knockout and variant-expression cell lines.
- **Limitations:** no established mammalian (mouse) genetic model in the reviewed literature; zebrafish morphants model neurodevelopmental but not the full multisystem phenotype.

Mechanistic Model — Integrated Narrative

The unifying lesion is loss of a single mitochondrial IMS quality-control enzyme. Structurally, Cupo et al. (2022) showed that human CLPB (Skd3) *"adopts a spiral hexameric arrangement that engages substrate via pore-loop interactions in the nucleotide-binding domain (NBD). Substrate-bound Skd3 hexamers stack head-to-head via unique, adaptable ankyrin-repeat domain (ANK)-mediated interactions to form dodecamers."* The decisive insight for inheritance is that *"SCN-linked subunits sharply inhibit disaggregase activity, whereas MGCA7-linked subunits do not. These advances illuminate Skd3 structure and mechanism, explain SCN and MGCA7 inheritance patterns."* [PMID: 36170828]

This single structural fact resolves the genotype–phenotype puzzle:

Allele configuration	Molecular effect	Clinical result
Biallelic LoF (stop/stop, severe missense)	No functional disaggregase	Full MEGCANN syndrome; ~66% mortality
Monoallelic missense near ATP pocket (SCN-type)	Poisons WT hexamer (dominant-negative)	Isolated severe congenital neutropenia, often no 3-MGA
Monoallelic null (heterozygous LoF)	Tolerated (pLI≈0)	Unaffected carrier

The zygosity-stratified 63-patient cohort empirically confirmed this gradient: *"Death occurred in 66% of biallelic and 23% of monoallelic individuals,"* and *"Structural predictions — particularly variants causing nonsense-mediated decay or ankyrin domain disruption — were stronger survival predictors than zygosity or age of onset."* [PMID: 41719910]

Evidence Base

PMID	Title (abbrev.)	Role
25597510	*CLPB mutations cause 3-MGA, brain atrophy, ID, neutropenia, cataracts, movement disorder*	Founding gene discovery; zebrafish causality; core phenotype
38270563	*CLPB disaggregase dysfunction impacts the SPY complex*	IMS disaggregase function; downstream SPY impairment
42244541	*HAX1 drives assembly and activation of CLPB*	HAX1 activation mechanism; metazoan conservation
41719910	*Zygosity-specific insights in 63 CLPB cases*	Genotype–outcome; mortality; prognostic predictors
34115842	*Heterozygous CLPB variants cause severe congenital neutropenia*	Dominant-negative monoallelic mechanism; hematopoietic model
34140661	*Neutropenia and ID are hallmarks of biallelic and de novo CLPB deficiency*	Phenotype enumeration; de novo dominant-negative
36170828	*Structural features of Skd3 disaggregase*	Hexamer/dodecamer architecture; inheritance explanation
39787888	*Untargeted metabolomics for 3-MGA syndromes*	3-MGA biochemistry; biomarker limitations
31729175	*Delayed appearance of 3-MGA-uria in neonates*	Diagnostic pitfall (fluctuant biomarker)
36485107	*Genetics of SCN as gateway to personalized therapy*	G-CSF/HSCT standard of care
37193639	*HAX1-related congenital neutropenia (SCNIR)*	G-CSF efficacy in partner-pathway disease
31030818	*Congenital neutropenia in children*	Treatment paradigm (G-CSF, HSCT)

Evidence source types: human clinical (25597510, 34140661, 41719910, 36485107, 37193639, 31030818, 39787888, 31729175); in-vitro/cellular (38270563, 34115842, 42244541); model organism (25597510 zebrafish); structural/computational (36170828, 42244541).

Limitations and Knowledge Gaps

- **Small patient numbers.** Even the largest cohort is 63 individuals; prevalence, penetrance, and natural-history estimates carry wide uncertainty.
- **Neurologic mechanism partly inferred.** The link from IMS proteostasis failure to progressive brain atrophy is strongly model-supported (zebrafish) but the human CNS cell-type-specific pathway is not fully demonstrated.
- **3-MGA pathogenesis.** The route to 3-MGA in secondary syndromes (acetyl-CoA buildup → trans-3-methylglutaconyl-CoA) is a mechanistic hypothesis, not fully proven for CLPB specifically; 3-MGA is a fluctuant, insensitive biomarker.
- **No mammalian (mouse) genetic model** is established in the reviewed literature, limiting preclinical therapy testing.
- **Cataract and movement-disorder mechanisms** at the tissue/cellular level remain under-characterized.
- **No therapeutic trials.** All treatment evidence is extrapolated from the SCN/HAX1 paradigm, not CLPB-specific trials.

Proposed Follow-up Experiments / Actions

1. **Generate a conditional mouse model** (tissue-specific *Clpb* knockout in hematopoietic and neural lineages) to test cell-autonomous mechanisms and serve as a preclinical platform.
2. **iPSC-derived neurons, cerebellar organoids, and granulocyte-differentiation assays** from patient cells to map cell-type-specific proteostatic collapse and screen small-molecule chaperone enhancers.
3. **Structure-guided therapeutics** exploiting the HAX1–CLPB activation axis — test whether boosting residual CLPB oligomerization/ATPase (or correcting dominant-negative poisoning) rescues disaggregase activity.
4. **Prospective natural-history registry** with standardized 3-MGA + plasma metabolomics, MRI, and neutrophil monitoring to validate the NMD/ankyrin-disruption prognostic predictor
(P 41719910).

5. **Systematic reclassification of the ~692 VUS in ClinVar** using functional disaggregase assays and structural prediction to improve diagnostic yield.
6. **Evaluate HSCT timing and outcomes** specifically in CLPB patients (vs generic SCN), given the multisystem neurologic burden transplantation does not address.

Report compiled from aggregated disease-level resources (OMIM, Orphanet, MONDO, gnomAD, ClinVar, UniProt, HGNC) and primary literature. All mechanistic and clinical claims are cited to PMID where available; ontology term suggestions (HPO, GO, CL, UBERON, CHEBI, NCIT) are provided for knowledge-base population.
